BIOT 6900 . Module 3 Assignment

Target Triage with Protein Language Models - Pancreatic Ductal Adenocarcinoma (PDAC)

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Date: October 14, 2026

1. Re-ranking

The final ranking generally agreed with the Module 2 ranking, as most genes changed by only 0 to 2 positions, indicating that incorporating tractability evidence caused a moderate reshuffling rather than a complete change in priorities. CEL showed the largest upward movement, rising from rank 11 to rank 6, a change of +5 positions. This movement was likely influenced by its high small-molecule tractability score (p_sm = 0.72), which strengthened its overall ranking despite its middling Module 2 position. I used a weight of w = 0.6, assigning 60% importance to Module 2 biological evidence and 40% to tractability. This weighting prioritizes biological relevance while still considering whether a target is feasible for drug development. Although a different weight could change the final rankings, this approach balances the strength of the biological evidence with the practical potential for therapeutic targeting.

2. Modality

I would choose ERBB4 because it shows a sharp difference between the two modalities, with p_sm = 0.70 and p_ab = 0.08. ERBB4 encodes a receptor tyrosine kinase in the ErbB family that is located on the cell membrane. When activated by its ligand, it triggers intracellular signaling pathways involved in cell growth, survival, and differentiation. Although part of the protein is exposed outside the cell and could potentially be targeted by an antibody, its intracellular kinase domain contains an ATP-binding pocket that small molecules can bind to and inhibit. This may explain why the model favors small molecules over antibodies for ERBB4.

3. Validation

	AUROC	Average precision
Random split	0.845	0.726
Family split	0.737	0.541

Chance level for average precision = 0.334.

The family-split result is more reliable because it tests whether the model can generalize to protein families it has not seen during training. In a random split, closely related proteins may appear in both the training and test sets, allowing the model to recognize familiar sequence patterns or structural features rather than learn generalizable relationships between proteins and druggability. This data leakage can inflate the apparent performance. The family split reduces this bias by keeping related proteins together in either training or testing. The decrease in AUROC from 0.845 to 0.737 and average precision from 0.726 to 0.541 suggests that the model performs less effectively on unfamiliar protein families. Although the family-split average precision remains above the chance level of 0.334, the results

indicate that generalization to new protein families is more challenging than the random-split results suggest.

4. Variants

Gene	Variant	Note	Score	Percentile
TP53	R175H	hotspot	-5.97	19.3
APOE	C130R	rs429358, defines e4 (C112R mature)	7.21	100.0
APOE	R176C	rs7412, defines e2 (R158C mature)	-8.56	11.8
TREM2	R47H	rs75932628, AD risk variant	-5.16	47.7

The variant analysis demonstrates that evolutionary surprise does not necessarily correspond to disease-causing potential. TP53 R175H had a score of -5.97 and a 19.3rd-percentile rank, suggesting that the substitution is relatively unusual to the model. APOE R176C (e2) had a score of -8.56 and ranked at the 11.8th percentile, making it one of the most evolutionarily surprising variants in this set. In contrast, APOE C130R (e4) had a positive score of +7.21 and ranked at the 100th percentile, indicating that the model considers the mutant amino acid highly evolutionarily plausible. However, APOE e4 is a well-established Alzheimer's disease risk variant, demonstrating that evolutionary plausibility does not necessarily indicate clinical safety. Similarly, TREM2 R47H had a score of -5.16 and ranked at the 47.7th percentile, placing it near the middle of the distribution despite its association with increased Alzheimer's disease risk. These examples highlight a key limitation of sequence-based models: they assess evolutionary patterns rather than directly measuring disease risk or functional consequences. They may not capture changes in lipid transport, amyloid-beta clearance, protein interactions, or downstream biological pathways. Therefore, variant scores should be interpreted alongside clinical evidence, functional experiments, and other genetic data before drawing conclusions about pathogenicity.

5. Limitations and Next Step

Sequence-based models can provide useful information about protein function, evolutionary conservation, and druggability, but they cannot fully determine a protein's three-dimensional structure, binding-pocket geometry, accessibility, or flexibility. Therefore, a high sequence-based tractability score does not necessarily mean that a protein has a suitable binding pocket or will be an effective drug target. For Week 4, I would select MUC5B and ERBB4 for structure-based pocket assessment because they represent two different cases requiring structural investigation. MUC5B ranks first with a final score of 0.880, a small-molecule tractability score of 0.045, and an antibody tractability score of 0.545. As a large, secreted mucin glycoprotein without a typical enzyme-like binding pocket, MUC5B would benefit from structural analysis to identify accessible regions that could potentially be targeted by antibodies. ERBB4 ranks second with a final score of 0.853, a small-molecule tractability score of 0.702, and an antibody tractability score of 0.083. Because ERBB4 is a receptor tyrosine kinase with an intracellular ATP-binding pocket, it is a plausible small-molecule target, but structural assessment is still needed to evaluate the pocket's geometry and suitability for drug binding. Comparing these proteins would help determine whether their sequence-based predictions translate into realistic structural opportunities for drug development. Finally, zero genes were unmapped in Step 1, so no additional evidence is needed to resolve unmapped targets.

Final Ranked Targets (Top 6 of 15)

Rank	Gene	week2_score	p_sm	p_ab	final
1	MUC5B	0.976	0.045	0.545	0.880
2	ERBB4	0.973	0.702	0.083	0.853
3	TNS4	0.975	0.214	0.003	0.733
4	COL11A1	0.984	0.096	0.000	0.707
5	FN1	0.971	0.477	0.292	0.653
6	CEL	0.954	0.717	0.009	0.600

Addendum: Disease and data choice; discordance and limitations

BIOT 6900 · Module 3 · Mohammed Sharukh. Added October 29, 2026, per the instructor's announcement asking every report to include these two sections. The analysis and numbers above are unchanged.

A. Disease and data choice, with biological interpretation

Disease and inputs. I kept pancreatic ductal adenocarcinoma (PDAC) from Module 2, because PDAC has few targeted therapies and its four drivers (KRAS, TP53, CDKN2A, SMAD4) are hard to drug. My input was the top 15 genes of `targets_pdac.csv`, built from open-access CPTAC-PDAC RNA, protein and somatic-mutation data (LinkedOmics). The tractability evidence came from the course data pack: ESM C 300M embeddings of the reviewed human proteome and about 1,200 panel proteins labelled by Open Targets clinical precedence. All 15 targets mapped (no non-coding genes), as expected, since the Module 2 join already required each gene to have protein data.

What the re-ranked targets mean.

- **MUC5B (new #1) and MUC5AC** were lifted by antibody priors (p_ab 0.545 and 0.592). Both are secreted gel-forming mucins that are aberrantly expressed in PDAC, so an antibody-style prior is biologically sensible. Mucins are already pursued as antibody and antibody–drug-conjugate targets.
- **ERBB4 (#2)** carries a strong small-molecule prior (p_sm 0.70) because it is a receptor tyrosine kinase with an ATP pocket. Its nearest drugged neighbours are other kinases, so this is a family argument, and its ErbB relative EGFR is already a drug target in pancreatic cancer.
- **CEL (+5, to #6)** has the highest small-molecule prior (p_sm 0.72), which fits an enzyme with a catalytic site. Biologically, however, CEL is an acinar digestive enzyme that is strongly *lost* in tumours (Module 2 RNA −6.09). It is tractable but not therapeutically relevant: "druggable" is not the same as "worth drugging".

Known genes recovered. The canonical drivers are not in the Module 2 top 15, so the prior never scored them. The top 15 do contain established PDAC-associated proteins: MUC16 (CA125), MUC5AC, and the stromal genes POSTN and FN1.

Surprising hit. TNS4 stayed #3 despite a low prior (p_sm 0.214, p_ab 0.003). It has the largest concordant tumour increase in the list, but it is a focal-adhesion scaffold with no close drugged relatives. It pairs strong biology with a tractability gap, which makes it a candidate for structure work or a non-classical modality (degrader or protein–protein interaction).

B. Discordance and limitations

Discordant gene: ERBB4. In Module 2, ERBB4 mRNA falls in tumours (log2FC −2.57) while its protein rises (+1.33). The Module 3 prior is sequence-based, so it scores the whole receptor kinase regardless of which form of the protein is actually present. If the protein signal comes from a cleaved fragment, from stroma, or from a stabilized but inactive pool, a high p_sm would

describe a drug-binding site that may not matter in tumour cells. ERBB4 therefore stays high, with a validation gate: confirm full-length receptor in tumour epithelium (for example by immunohistochemistry).

Discordance between evidence types. POSTN is biologically an ideal antibody target (secreted, stromal, concordant up), yet its antibody prior is only 0.048, dropping it to 11th. A likely reason is that the panel labels clinical *precedence*, and few extracellular-matrix proteins have approved antibodies. The prior therefore encodes drug-development history, not accessibility. Conversely, CEL scores high on the prior but low on biology.

What I can claim. The prior shows which of my targets resemble already-drugged proteins, and by which modality, with moderate ability to generalize to unseen families: family-split AUROC 0.737 and AP 0.541, against a chance AP of 0.334.

What I cannot claim. I cannot claim that any target is actually druggable, because sequence shows no pocket geometry; that a high probability predicts efficacy; or the right direction of intervention (PLIN4 and NOS1 are *decreased* in tumours, so inhibiting them has no rationale). Random-split metrics are inflated by homology, and clinical-precedence labels carry historical bias. The anchor-variant scores measure evolutionary surprise, not pathogenicity.

Note on the Week 4 hand-off. This report proposed MUC5B and ERBB4 for structure assessment. In Module 4, MUC5B (5,762 residues) exceeded the AlphaFold Database's 2,700-residue limit, so I used CEL, the highest small-molecule prior, instead.